NOT MEASUREMENT SENSITIVE

MIL-T-62276C
17 July 1991
SUPERSEDING
MIL-T-62276B
11 February 1987

MILITARY SPECIFICATION

TRUCK TRACTOR: DIESEL ENGINE DRIVEN, 80,000 POUNDS GVW, 145,000 POUNDS GCW, 8X4, MODIFIED COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tilt cab, diesel engine driven, eight-wheel, four-wheel drive truck tractor having a minimum gross vehicle weight (GVW) of 80,000 pounds and a minimum gross combination weight (GCW) of 145,000 pounds, for normal operating conditions. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specification, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2320

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SPECIFICATIONS

FEDERAL

	A-A-393	- Extinguishers, Fire, Dry-Chemical (Hand Portable).
	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
*	VV-F-800	- Fuel Oil, Diesel.

MILITARY

*	MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
	MS 51118	- Pintle Assembly, Towing - 40,000 Lbs. Capacity, Manual Release.

* (Unless otherwise specified, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF DEFENSE (DOD)

Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-6093.)

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations.

Federal Motor Vehicle Safety Standards.

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(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
* Noise Emission Standards for Transportation Equipment Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practices

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply), Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
J537	- Storage Batteries.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J589	- Turn Signal Switch.
J682	- Rear Wheel Splash and Stone Throw Protection (DOD adopted).
J688	- Truck Ability Prediction Procedure (DOD adopted).

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J700	- Fifth Wheal Kingpin - Commercial Trailers and Semitrailers (DOD adopted).
J844	- Nometallic Air Brake System Tubing (DOD adopted).
J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to the SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF WASHINGTON
Overweight/Oversize Vehicle Permits

(Application for copies of State of Washington publications should be addressed to the Washington State Department of Transportation, Olympia, WA 98504.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

* (Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 176 Montrose West Ave., Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.8, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.8, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical

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data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available in the engineering offices of the procuring activity prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.1.1 Marking and data plates. As specified by the procuring activity for the appropriate military service (6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

3.1.1.2 Rustproofing. The vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.3 Drain plugs. Drain plugs installed in the transmission and rear axles shall be of the permanent magnet type.

* 3.1.1.4 Towing devices. Towing devices consisting of two hooks, loops, eyes or pins or the chassis manufacturer's standard single center mounted eye or pin shall be mounted on the front of the vehicle. When specified (see 6.2), in addition, towing devices shall be mounted on the rear of the vehicle. All towing devices shall be frame rail mounted or reinforced back to each frame rail.

* 3.1.1.5 Wheel splash and stone throw protection. The vehicle shall have rigid quarter-fenders to the front of the rear wheels and rubber mud flaps to the rear of the rear wheels. Mud flaps and their extension supports shall be readily removable, to increase landing wheel clearance, without the use of hand tools. The quarter-fenders need extend down only to the height of the centerline of the rear axle. Cab shall have rubber mud flaps to the rear of the front wheels. All splash shield and mud flap installations, front and rear, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682. A metal strip not less than .125 inch thick and not less than 1 inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1.25 wide by 0.094 inch thick shall be furnished at each bolt.

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3.1.1.6 Trailer towing package. A trailer towing package consisting of a pintle, safety chain attachment devices, a lighting receptacle, a trailer brake control system, and associated reinforcements and wiring shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS51118. The pintle shall be installed on the chassis frame with reinforcements to transfer a vertical tongue load of not less than 4,000 pounds and a horizontal drawbar load of not less than 40,000 pounds directly to the chassis frame rails. The rearmost portion of the pintle shall be forward but not more than 4 inches forward of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength of not less than 40,000 pounds. The attachment devices shall be capable of accommodating a standard grab hook (4-9/16 inches wide, 1-3/16 inches thick, 25/32-inch throat width) for a 5/8-inch chain. The lighting receptacle, conforming to SAE J560 with its conductors connected and color-coded as specified therein, shall be mounted in a readily accessible location near the pintle. The trailer brake control system shall conform to 3.4.11.3.

3.1.1.7 Spark arrester. As specified herein (see 3.4.4), the vehicle shall be furnished with an exhaust system spark arrester.

* 3.1.1.8 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with the EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The interior sound level shall not exceed 84 db(A), when measured in accordance with Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to the EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

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3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; fifth wheel; and full complement of fuel, lubricants and coolant. The curb weight shall be not greater than 22,000 pounds.

3.2.5 Gross vehicle weight. GVW shall consist of curb weight, operator (weight computed at 175 pounds), and a payload to provide not less than the specified GVW.

3.2.6 Weight distribution. The suspension system, axle and tire capacities shall be not less than 18,000 pounds on the front axle and 46,000 pounds on the rear bogie. The fourth axle shall be a spring up, air pressure down, self-steering, pusher type (dead) axle located forward of the rear bogie, as specified in 3.4.9.3. The fourth axle shall have suspension, axle and tire capacities of not less than 20,000 pounds. The rear bogie and the fourth axle combined shall be capable of supporting a load of not less than 65,000 pounds with the vehicle operating at 55 miles per hour (mph).

3.2.7 Gross combination weight. GCW shall consist of the truck tractor curb weight, operator weight (computed at 175 pounds), and the weight of a semitrailer loaded to provide not less than the specified GCW. The fifth wheel shall be located so that with the truck tractor loaded to GVW, the load ratings of the chassis components are not exceeded.

3.2.8 Rating. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the engineering office of the procuring activity. The minimum GVW rating shall be 80,000 pounds. The minimum GCW rating shall be 145,000 pounds.

3.2.9 Washington State vehicle laws. The vehicle, when loaded to rated GVW, shall comply with all State of Washington motor vehicle laws in effect on the date of manufacture, including regulations and conditions governing "Overweight/Oversize Vehicle Permits."

3.2.10 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.11 Asbestos. Asbestos materials shall not be used in any form in any part of the vehicle.

* 3.2.12 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

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3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the vehicle loaded to the specified GCW.

3.3.1.1 High speed gradeabilitiy. The vehicle shall ascend a continuous grade of 0.22 percent at 50 mph. Gradeability requirements shall be met with the main transmission in direct drive, and with the axles in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed gradeability. The vehicle shall ascend a continuous grade of 7.0 percent at 5 mph. The vehicle shall be capable of making a complete stop while ascending a 7.0 percent grade and accelerating from stopped position to 5 mph up the 7.0 percent grade. Gradeability requirements shall be met with the transmission and axle in low speed range. Gradeability shall be verified with calculations in accordance with SAE J688.

3.3.1.3 Low speed. Low speed shall be calculated in forward and in reverse with the engine operating at not less than 35 percent of recommended governed speed, and shall provide a vehicle speed of not more than 10 feet per minute. In lieu of calculations, low speed requirement may be demonstrated with driver assistance. The vehicle shall be capable of sustaining a load of 87,000 pounds at the fifth wheel, gradually reducing to 54,200 pounds at the fifth wheel while the tractor-semitrailer combination is moving at 10 feet per minute over a distance of 40 feet. Loads shall be sustained without the permanent deformation or failure of any part and without excessive deflection.

3.3.1.4 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 55 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3).}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the tractor-semitrailer combination, loaded to specified GCW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

* 3.3.2.1 Emergency brakes. The emergency brakes shall stop the tractor-semitrailer combination, loaded to specify GCW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52(b).

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3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. The vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than six cylinders. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating JP-4, JP-5 or JP-8 is acceptable.

3.4.1.2 Oil filter. A full-flow or combined full-flow and bypass oil filter with replacement element shall be furnished.

* 3.4.1.2.1 Oil cooler. An engine oil cooling system shall be furnished.

* 3.4.1.3 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

* 3.4.1.4 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than two-quart capacity.

3.4.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. The control shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters, or complete thermostatic control of all coolant flow through the radiator. On tilt cab models, a radiator servicing access door shall be provided to allow verification of the coolant level.

* 3.4.1.6 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

* 3.4.1.7 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and a fuel warmer shall be provided. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block, including a fuse or circuit breaker, to a single three-pronged (male), weatherproof, slave receptacle for receiving external

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power and grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. Connecting cable shall include a matching female connector at the vehicle end and a standard, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60 °F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provision for storage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

(a) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. An engine thermostat with an operating range of 170 °F to 195 °F shall be installed;

* (b) An oil pan heater of the permanent external surface mount, permanent in-pan mount, or immersion type that meets the following requirements, shall be installed:
  (1) Not less than 22 watts per quart heating capacity
  (2) Not more than 18 watts per square inch heating capacity
  (3) Thermal balance design or thermostat control providing for uninterrupted operation
  (4) Provision for mounting below minimum service oil level.

(c) A fuel warmer or pre-heater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 90 °F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet inside of the fuel warmer unit.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. A 12- or 24-volt starting system, with a 12-volt lighting system shall be furnished. Engine starting equipment include an ether starting system or glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

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3.4.2.2 Alternator. Unless otherwise specified (see 6.2), a minimum 85 ampere alternator shall be furnished. The alternator output with the engine at idle speed shall be not less than 60 amperes.

3.4.2.3 Lighting. All vehicle lights, reflectors and wiring shall be in accordance with Federal Motor Carrier Safety Regulations 393.13, 393.19, 393.20 and 393.22 through 396.26(d). Lights and reflectors shall not be mounted on the vehicle bumpers (unless recessed and fully protected).

* 3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Turn signals units shall be installed in accordance with SAE J588. Turn signal units shall be visible and audible in combination with the towed vehicle and shall be visible when not in combination with the towed vehicle. Turn signal operating units shall have visible and audible type flash indicators.

3.4.2.5 Semitrailer lighting cable. The semitrailer lighting cable shall conform to SAE J1067 and shall incorporate a connector conforming to SAE J560 on the semitrailer end. The cable shall be coiled and shall have a length of not less than 180 inches when fully extended. The SAE J560 connector shall include a grip for withdrawing from the semitrailer receptacle. Stowage for the cable shall be provided by means of a hook and hanging loop or protective holding bracket. When the hook and loop method is used, the cable shall be furnished with a hanging loop, so attached as to insure that the plug is pointed down when the jumper cable is stowed in the hook provided on the rear of the truck tractor cab or on a pogo stick type hose tender. A protective bracket, when furnished, shall be mounted at the rear of the cab, below the roof line, and shall hold the cable plug within the bracket housing to prevent entrance of weather into the terminals.

3.4.2.6 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating shall be not less than 500 minutes and the total cold cranking rating at 0°F shall be not less than 1,740 amperes. Reserve capacity and cold cranking ratings shall be in accordance with SAE J537. Batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

3.4.2.7 Fog lights. Fog lights, one on each side of the front of the vehicle, shall be furnished.

3.4.2.8 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

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3.4.3.1 Air cleaner. An air cleaner shall be furnished.

3.4.3.2 Fuel tank(s). The vehicle shall be equipped with fuel tank(s) of not less than 75 gallons total capacity. In addition, fuel tank capacity shall exceed 75 gallons if required to provide for not less than 200 miles operating range with the vehicle loaded to specified GCW. When more than one tank is furnished, means shall be provided to assure an equalized fuel level in both tanks.

3.4.3.3 Fuel filters. Fuel filters shall be furnished. Primary and secondary fuel filters shall be furnished when the manufacturer's standard.

* 3.4.3.4 Fuel and water separator. When specified (see 6.2), in addition to the manufacturer's standard fuel filters, a fuel and water separator shall be furnished. The separator shall include a water coalescer and a drain valve. When a separator is required, a combination filter/separator unit may be furnished.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. The tailpipe shall be vertically mounted at the rear of the cab and shall be furnished with a heat shield and a hinged rain cap. The tailpipe shall extend approximately 12 inches above the top of the cab, but not exceeding a vehicle height of 150 inches. Unless otherwise specified (see 6.2), a spark arrester having a 80 percent arresting efficiency when rated in accordance with SAE J350 shall be furnished, except on turbocharged engines. When specified (see 6.2), a spark arrester shall be furnished on turbocharged engines to complement the spark arresting qualities of the turbocharger. Maximum exhaust system restriction, as established by the engine manufacturer, shall not be exceeded.

3.4.4.1 Pyrometer (s). The vehicle shall be provided with a pyrometer in the engine exhaust manifold. When a "V" type engine is furnished, a pyrometer shall be furnished in each exhaust manifold. When a turbocharged engine is furnished, an additional pyrometer shall be installed on the exhaust side of the turbocharger. When two or three pyrometers are furnished, the gages shall be incorporated in one housing. The gage(s) shall be mounted on the instrument panel.

* 3.4.5 Automatic transmission. The vehicle shall be provided with an automatic transmission. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall include a hydraulic torque converter and not less than five forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with a power takeoff opening. A transmission oil cooler system shall be provided. The transmission shall be provided with a built-in hydraulic retarder.

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3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be a reinforced or heavy duty type, capable of supporting the loads specified in 3.2.8 and 3.3.1.3. Reinforcement, when provided, shall extend at least from the rear of the front suspension rear hanger bracket to the bogie trunnion mounting bracket. A heavy duty frame shall have frame rails of greater section modulus than the manufacturer's standard for the class vehicle furnished and shall provide structural strength at least equivalent to the reinforced frame.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to that specified in 3.2.6 and to the load imposed on each member, measured at the ground, with the vehicle loaded to specified GVW. When spring capacity is rated at the spring pads, unsprung weight shall be deducted.

3.4.9 Axles. Axle ratings shall be at least equal to those specified in 3.2.6 and to the load imposed on each axle, measured at the ground, with the vehicle loaded to specified GVW.

3.4.9.1 Rear and fourth axle Lubrication. The front and fourth axle wheal bearings shall be oil-lubricated. The oil type and viscosity shall be in accordance with the manufacturer's recommendations. The hub caps shall have windows for visual determination of the oil level. Provisions for venting and replenishing the oil supply shall be incorporated. The hubs shall be fitted with seals.

3.4.9.2 Rear bogie. The rear bogie shall be of the four-wheel type, complete with axles, springs, torque rods and all other necessary parts. The bogie shall be provided with means for permitting differential action between the two axles. A manually or automatically controlled lockout, assuring equal power to each rear axle, shall be provided. The manual lockout control used shall be located in the cab and shall be provided with a guard to prevent accidental actuation. A warning light for the lockout control shall be provided on the instrument panel. The rear bogie axle spacing, measured from the centerline of the forward rear axle to the centerline of the rear axle, shall be not less than 51 inches and not more than 55 inches. The rear bogie axles shall be of the one- or two-speed type, with the two-speed type equipped with electric, vacuum or air shift. Axles shall be provided with gear ratio(s) that will permit proper gear selection and the performance specified in 3.3.1 through 3.3.1.4. A temperature gage shall be provided for each bogie axle and shall be mounted on the instrument panel.

3.4.9.3 Fourth axle. The fourth axle shall be a spring up, air pressure down, self-steering, pusher type (dead) axle, mounted forward of the rear bogie. A lockout for the self-steering portion of the pusher axle shall be provided. With the fourth axle in the lowered position, the

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dimension, measured from the centerline of the fourth axle to the centerline of the rearmost axle of the bogie assembly, shall be not less than 121 inches and not more than 124 inches.

* 3.4.10 Wheels, rims, tires and tubes. The vehicle shall be equipped with single front wheels and dual wheels on both the rear bogie and the fourth axle. Rim and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations, for the type and size of tires furnished. Wheels of the bogie and fourth axles shall be interchangeable without the use of an adapter. Ten-stud disc type wheels shall be furnished.

* 3.4.10.1 Tires. Unless otherwise specified (see 6.2), tires shall be of the tubeless type with highway tread. Unless otherwise specified, low profile or standard profile tires may be furnished. Except as specified in 3.2.6 tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded to specified GVW. Tire surface pressure shall not exceed 100 pounds per square inch (psi). Tires and tires size designation systems shall conform to the Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

3.4.10.2 Inner tubes. For tube tires, inner tubes shall be of the heavy duty type and shall be of proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

* 3.4.10.3 Carrier for spare tire assembly. When a spare wheel or rim or a spare tire assembly is specified (see 3.4.10.4 and 3.4.10.5), a carrier shall be furnished. The carrier shall be mounted behind the cab, above the chassis frame. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of, or plated with, corrosion-resistant material. When two different size wheels are furnished on the truck tractor, one carrier for each size shall be furnished. The carrier design shall enable safe removal or mounting of a spare wheel assembly using only the tools specified in 3.4.16.1. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. If two different size wheels are furnished on the truck tractor, one wheel of each size shall be furnished. When a spare tire assembly is specified (see 3.4.10.5), a spare wheel or rim shall be furnished.

3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare

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wheel or rim. If two different size tire assemblies are furnished on the truck tractor, one tire assembly of each size, tread design and load range (ply rating) shall be furnished.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42(b), 393.43, and 393.45 through 393.52.

3.4.11.1. Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air- or water-cooled, and having a capacity of not less than 14-1/2 cubic feet per minute (cfm)
(b) Air storage reservoir(s) with not less than 6,000 cubic inches total capacity, each tank equipped with a drain, and with safety and check valves between the compressor and the last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low air pressure warning, visible and audible
(g) Service brake stop lamp twitch
(h) Automatic moisture ejector
(i) Parking (emergency) brakes on both axles of the rear bogie
(j) Automatic slack adjusters.

3.4.11.1.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.2 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brakes are furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

Trailer brake control system. In addition to the components specified in 3.4.11.1 and 3.4.11.1.1, a trailer brake control system shall be furnished. The trailer brake control system shall include:

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(a) Identification of emergency and service lines
(b) Coincident control of trailer brakes with prime mover foot control
(c) Independent hand control for trailer brakes
(d) Prime mover protection valve with dash control and automatic breakaway feature
(e) Trailer stoplight control operable with foot brake and with hand control for trailer brakes
(f) Two SAE J844 coiled connecting air hoses, not less than 180 inches in extended length, equipped with coiled spring hose guards, and SAE J318 gladhand quick connector on trailer end of hoses
(g) Spring supports on truck tractor cab or on a pogo stick type hose tender with dummy gladhand connectors to retain hoses when not in use. Supports shall not be mounted on the cab roof
(h) Prime mover only parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system.

3.4.12 Cab. A nonsleeper, tilt type cab shall be furnished. The bumper to back-of-cab dimension shall be 50 to 64 inches. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of the Federal Motor Carrier Safety Regulation 399, including the first step height, which shall be not more than 24 inches. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least the curbside door equipped with external, key-operated lock. The cab shall have upholstered, individual, adjustable driver's seat and individual passenger seat. The driver's seat shall be the manufacturer's standard air ride type. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Safety grips shall be provided on each side of the cab to assist personnel entering or leaving the cab and, in addition, to assist personnel in climbing onto the truck tractor deck plate. Interior lighting shall be provided. Two pairs of seat belts shall be installed. Drip rails shall be installed above the cab doors.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type operated by air or electric motor(s).

3.4.15 Bumper. Unless the bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided.

3.4.16 Tool stowage. Stowage space of sufficient size to accommodate the vehicle jack, hand tools, antiskid chains and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. The stowage space shall provide for positive retainment of this equipment during vehicle operation. Stowage space may be furnished inside the cab. When stowage space is located outside the cab, it shall be weatherproof and shall provide for looking with a padlock.

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3.4.16.1 Tools. When specified (see 6.2), each vehicle shall be furnished with tools required for exchanging the mounted tire assembly with the spare assembly (see 3.4.10.3), and shall include at least a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under an axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly.

3.4.16.2 Emergency reflective triangles. Three emergency reflective triangles conforming to Federal Motor Carrier Safety Regulation 393.95(f)(2)(i) shall be furnished and properly stowed.

3.4.16.3 Fire extinguisher. One 5-pound chemical type fire extinguisher shall be mounted in a convenient location in the cab. The fire extinguisher shall have an Underwriter's Laboratory rating of 10 B:C or more, shall include a metal safety pin and wire, and shall conform to type I, class 2, size 5 of A-A-393. The extinguisher mounting device shall be of the quick release type. The fire extinguisher shall permit visual determination of condition of charge.

3.4.17 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to the floor and to the defroster louvers. The heater shall be complete with a blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. The controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver’s compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing.

* 3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated starting switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or indicator
(e) Engine temperature gage or high coolant temperature or low coolant level red indicator warning light
(f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
(h) Tachometer

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(i) Ash receptacle
(j) Dual sunvisors
(k) Driver's compartment ventilator other than window
(l) Transmission temperature gage
(m) Temperature gage for each axle of the bogie
(n) Vernier type hand throttle
(o) A warning light and buzzer for low oil pressure
(p) Engine pyrometer gage(s).

* 3.4.20 Rearview mirrors. Outside rearview mirrors shall have not less than two supporting arms and shall be mounted on each side of the cab. The mirror support arms shall be adjustable to provide rearward vision when the vehicle is towing a 10-foot wide semitrailer. Flat and convex mirrors shall be furnished. At least 50 square inches of flat reflective area and a convex surface having at least 24 square inches of reflective area shall be provided with a radius of curvature of not less than 20 inches. Both flat, mirrors shall be of the self-contained, heated type controlled by a switch in the driver's compartment.

3.4.21 Horn. Manufacturer's standard electric horn shall be furnished. In addition, an air operated horn shall be furnished.

3.4.22 Engine hour meter. When specified (see 6.2), an engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4.23 Bank-up alarm. When specified (see 6.2), the vehicle shall be furnished with an audible, pulsating, signaling device (electrical or mechanical) to caution personnel when the vehicle is in reverse gear operation.

* 3.4.24 AM/FM radio. When specified (see 6.2), the manufacturer's standard AM/FM radio shall be provided.

3.5 Body. The rear-of-cab to the centerline of the rear bogie dimension shall be 160 inches, plus or minus one inch. The tractor shall be equipped with a sliding, fore and aft rocking, 36-inch diameter, air looking type fifth wheel, for SAE J700 kingpin. The fifth wheel shall have a sliding range of not less than 47.5 inches in increments of not less than 1.50 inches and not more than 4 inches.

3.5.1 Fifth wheel location. Fifth wheel sliding range shall be not less than 45 inches forward of the centerline of the bogie, and 2.5 inches, plus 0, minus one inch, rearward of the centerline of the bogie. With the fifth wheel in its most forward position, the clearance from the centerline of

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the kingpin to the cab, or to the vertical spare tire assembly, or to the pogo stick hose tender when furnished, shall be not less than 85 inches. With the fifth wheel in its most forward position, the landing wheel clearance from the centerline of the kingpin to the rear tires and chassis frame rails shall be not more than 106 inches. The vehicle shall be capable of mating with the semitrailers shown in figure 1.

3.5.2 Mounting. Fifth wheel mounting shall conform to Federal Motor Carrier Safety Regulation 393.70(b).

3.5.3 Fifth wheel height. The unladen level height of the fifth wheel shall be 53 inches, plus or minus one inch, above ground level.

3.5.4 Deck plate. Except for vehicles furnished with saddle tanks, a self-cleaning grating of sufficient structural strength for operator use in connecting air and electric lines between the truck tractor and a semitrailer shall be installed. The grating shall extend across and shall be bolted to the frame rails with sufficient overhang to cover and protect side mounted tanks and fittings. The grating shall be located as close to the cab as possible and shall extend not less than 36 inches towards the rear of the vehicle. Access through the grating shall be furnished for maintenance of fittings and other equipment. Provisions shall be furnished to allow access for personnel climbing onto the deck plate.

3.5.5 Hose tender. A pogo stick type hose tender shall be provided, mounted rearward of the spare tire carrier(s). The pogo stick shall be mounted on the vehicle deck plate or chassis members.

3.5.6 Wind deflector. When specified (see 6.2), a wind deflector shall be furnished with the vehicle for subsequent installation on the cab roof by the receiving activity, or shall be factory installed. The deflector shall be of molded fiberglass reinforced plastic, shall be not less than 63 inches wide, and shall be of a height suitable for use with the vehicle cab furnished and semitrailer vans having a level height of 12 feet 6 inches at an upper fifth wheel height of 48 inches. The deflector, including all mounting and supporting hardware, support ribs and installation instructions, shall be installed or securely stowed on the vehicle for shipment.

3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following alinement of lights; adjustment of engine; adjustment of brake system; charging of batteries; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine, running gear, and mounted equipment with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives.

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3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

4 QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor’s inspection are being properly applied. Failure of the contractor to promptly correct product’s deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First product vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine the curb weight and distribution of the curb weight on the front, fourth, and rear axles. The total imposed loading on the front, fourth, and rear axles shall be computed by the contractor and verified by the Government, using the curb weight, the operator weight at 175 pounds, and the payload required

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to provide the specified GVW. Except as specified in 3.2.6.1, the calculated imposed loads on front, fourth, and rear axles shall be compared to the suspension, axle, and tire capacity ratings to determine if those components are of adequate capacity to meet contractual requirements.

4.3.2 Road test. The first production vehicle shall be examined and road tested with payload, by the contractor, to assure that the vehicle will operate in accordance with contractual requirements.

4.3.3 Static load test. The first production vehicle shall be subjected to an 87,000 pound load on the fifth wheel. The vehicle shall be operated, with the 87,000 pound load, in forward and reverse directions, over a distance of 40 feet at a speed of 10 feet per minute. At the end of this test there shall be no evidence of failure or permanent deformation of any truck tractor component.

4.3.4 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.5 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that each vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements specified herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, his records indicating that the servicing and adjusting required by 3.6 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicle covered by this specification is intended for use by the Government as a prime mover for missile storage-erection semitrailers.

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* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
* (b) Issues of DODISS to be cited in the solicitation and if required, the specific issues of individual documents referenced (see 2.1.1 and 2.2.).
(c) Identification of appropriate service for painting and marking (see 3.1.1.1).
(d) Concealed markings, if required (see 3.1.1.1).
(e) Tropical rustproofing, if required (see 3.1.1.2).
(f) Power plant heaters and fuel warmer, if required (see 3.4.1.7).
(g) If a battery heater is not required (see 3.4.1.7).
* (h) Other than an 85 ampere alternator, if required (see 3.4.2.1.1).
(i) Fuel and water separator, if required (see 3.4.3.4).
(j) If a spark arrester is not required (see 3.4.4).
(k) Spark arrester, if required on turbocharged engine (see 3.4.4).
(l) Low profile tires, if required (see 3.4.10.1).
(m) Spare wheel(s) or rim(s), if required (see 3.4.10.4).
(n) Spare tire assembly(s), if required (see 3.4.10.5).
(o) Brake control for use from a towing vehicle, if required (see 3.4.11.2).
(p) Tools, if required (see 3.4.16.1).
(q) Engine hour meter, if required (see 3.4.22).
* (r) AM/FM radio, if required (see 3.4.24).
(s) Back-up alarm, if required (see 3.4.23).
(t) Wind deflector, if required (see 3.5.6).

* 6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. The SAE Work Sheet Item 1 should include vehicle model number, engine model number, and vehicle type, class and model. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still dry air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. 0.225
Table 5	- Velocity Factor. 250.0
Table 6	- Altitude Factor. 1.00

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Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 0.75
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0

* 6.4 Subject term (key word) listing.

Fourth axle
Heavy loads
Pusher axle
Truck, nontactical.

6.5 Identification of changes. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Preparing activity:
Army - AT

Project No. 2320-0573

Review activity:
Air Force - 84

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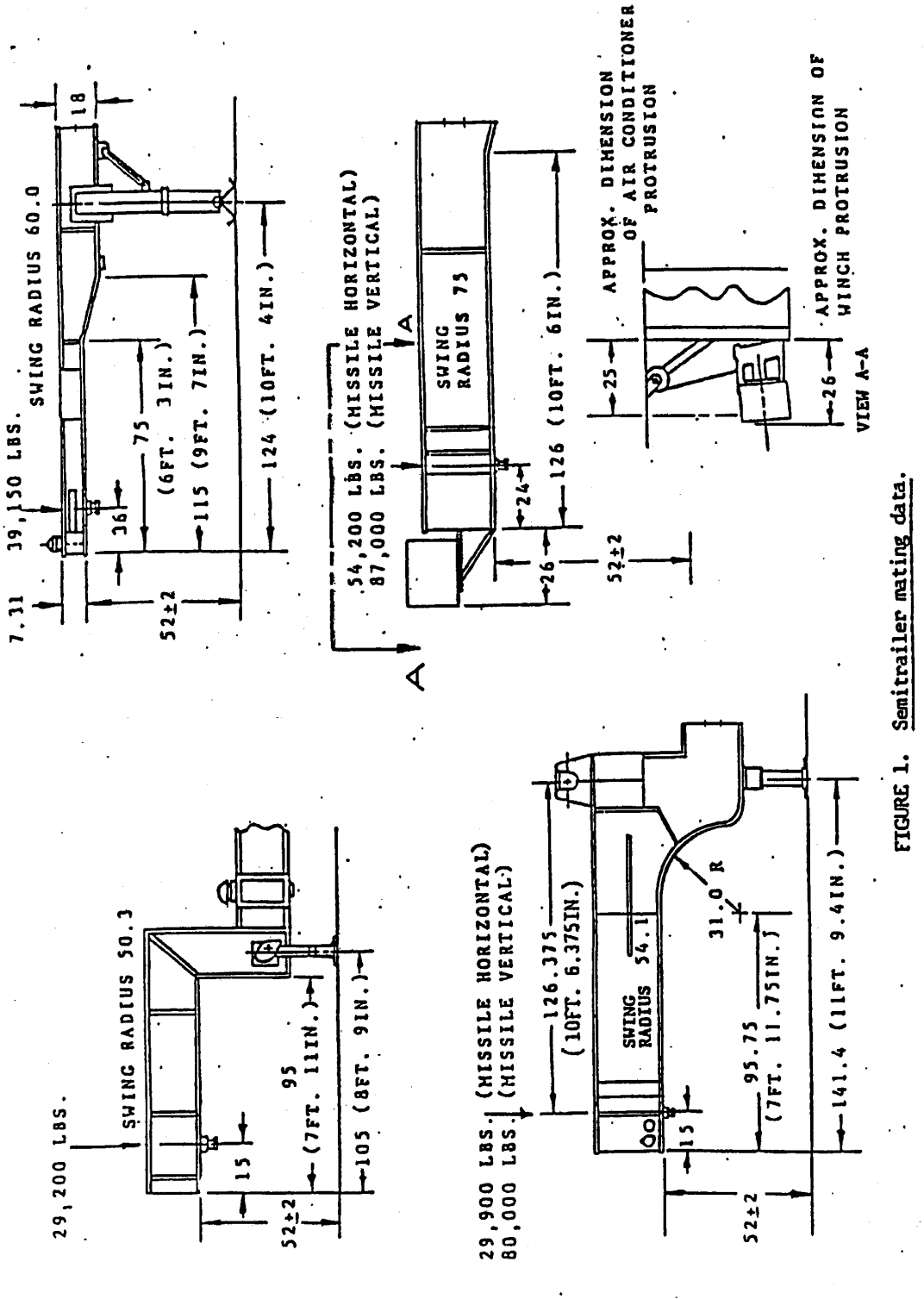


FIGURE 1. Semitrailer mating data.

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